SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

XGB PLC CPU 16 I/O (PLC-XGB-XBM-DR16S) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 50 mm above and below the base for ventilation; keep 100 mm from power cables
- Operating temperature: 0 to +55 °C
- Degree of protection: IP20
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check power supply and base grounding (FG)
2. Download I/O parameter and program
3. Verify each input and output point
4. Check communication settings
5. Back up the final project

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
CPU stops with watchdog error	Scan time exceeds watchdog due to program loop	TG-PLC-XGB-XBM-DR16S §3.1
Random watchdog error in noisy environment	Electrical noise due to improper grounding	TG-PLC-XGB-XBM-DR16S §3.2
I/O configuration mismatch after module replacement	Replacement module type differs from I/O parameter	TG-PLC-XGB-XBM-DR16S §3.3
Battery low warning	Backup battery end of life	TG-PLC-XGB-XBM-DR16S §3.4

On-site commissioning support can be requested through the Global CS Center. For programmable logic controllers the request is handled by the technical support team.